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Systematic review

Risk Mitigation in CI/CD Pipelines for Safety-Critical Embedded Systems Using Automated Testing and Rollback Strategies: A systematic literature review

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Abstract

The principles of traditional CI/CD pipelines are not suitable for safety-critical systems due to complexity safety regulation compliance. Therefore, benefits, like automated testing and fail-safe mechanisms, of CI/CD pipelines are explored and determined how it can be implemented to these embedded systems while handling the challenges associated with it.

Keywords: Continuous Integration and Continuous Development (CI/CD); Explainable Artificial Intelligence (XAI)

1. Introduction

The delivery of software has evolved from timed releases to deployments that can occur at any time, where Continuous Integration and Continuous Development (CI/CD) pipelines are utilised to enable this functionality while also providing additional benefits to software development efficiency through automated testing and fail-safe mechanisms (Vasiliev, James Morrison and Zhang, 2025). However, adopting CI/CD in safety-critical embedded systems introduces unique challenges such as complying with safety regulatory standards as well as the complexity associated with embedded systems due to the various hardware, software and computational components (Widyastomo and Lefaan, 2025). Consequently, any issues that remain undetected in the CI/CD process create the possibility where people or the environment are harmed due to the nature of safety-critical environments like healthcare or autonomous vehicles. Therefore, rigorous testing procedures to detect issues and fail-safe mechanisms to ensure that a system remains in a safe state is imperative (Sundara Raju, Yie Leong and Kumar Subramaniam, 2026).

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2. Materials and Methods

This systematic literature review aims to evaluate automated Continuous Integration and Continuous Development (CI/CD) techniques for safety-critical embedded systems by utilising seven research papers to achieve the following research objectives:

- **RO1:** To identify key risks associated with automated deployments in safety-critical systems.
- **RO2:** To examine automated testing techniques suitable for embedded CI/CD pipelines.
- **RO3:** To design a risk-aware CI/CD pipeline incorporating rollback and fail-safe mechanisms.

Moreover, utilising the findings found when achieving these research objectives will enable the following research question to be answered:

- **RQ:** How can automated testing and rollback strategies within CI/CD pipelines reduce deployment risks in safety-critical embedded systems?

3. Literature review

This section provides a detailed analysis of all seven research papers including how each are structured, the type of methodology that it follows, how each paper contributes to the research objectives as well as any research gaps and comparisons that are identified.

3.1. Common research paper section identification and analysis

Commonly occurring sections that appear in most of the seven research papers include the following:

- **Abstract:** A section that highlights information that best describes the content of the research paper as a whole, enabling researchers to quickly obtain a general understanding of the paper. The key information that an abstract is likely to contain is the context that the paper is set in, such as the current issues related to the topic, as well as the type of contributions that the paper's researchers provide, like a developed solution or a conclusion drawn from other sources. This section is written in a concise style, because the overall content of the paper is shortened into a summarised format.
- **Introduction:** An introduction in a research paper is similar to the abstract where it describes the content of the research paper. However, it provides a more in-depth explanation in comparison to an abstract and primarily focuses on the context of its topic, including current challenges, gaps, recent activity, and how the paper contributes to the topic further. It utilises a persuasive writing style to indicate to the reader how the paper's contributions can influence the challenges or gaps that were identified.
- **Background:** This section in a research paper further expands on the details surrounding the context of the paper's topic as well as additional concepts that weren't indicated in the introduction, including work done by other researchers that is a consideration for the paper. This section also uses a persuasive writing style to further showcase the impact that the paper's contributions can have on the context.

- **Methodology/Research method:** This section the methodology that the paper's research team follows when creating their research paper, which defines what kind of data they collect, how they collect it and how they will utilise it in the research paper. It utilises an objective writing style as it clarifies how the research is conducted as well as what data is collected and how it will be utilised to achieve a result.
- **Literature review/Related work:** This section describes and analyses the content from similar research papers and how these papers influenced the main paper. The content found in this section depends on what the paper is attempting to achieve such as utilising the statements and outcomes from other papers to justify a conclusion or utilising frameworks from other developed solutions to develop a new solution. The writing style can be objective if it's a framework where there is data that proves it's effectiveness or persuasive if the information from the other papers is used to reach a consensus.
- **Proposed architecture:** This section describes a framework or system that the paper's research team has developed for the paper, which can include system's physical components, how it functions and makes decisions on its own, how it complies with regulations as well as how it collects, analyses and stores data. This section is objective as the research team, because the paper's research team are the ones who developed the system and that would make them experts on how it functions.
- **Challenges:** This section indicates in detail the challenges associated with the context of the paper's topic or the solution suggested by the paper that have to be considered. It utilises a persuasive writing style as the topic challenges can be used as reason to promote the paper's solution or mitigations if it's challenges associated with the solution.
- **Limitations:** This section indicates the limitations and gaps of the paper itself that the paper's researchers consider to be impactful and should be considered, such as the amount and variety of data collected. This section is critical as potential flaws that can impact the result of the paper are specified.
- **Case studies:** This section identifies real-world examples that act as evidence to support arguments made in the paper, like the outcome when an organisation utilises the solution proposed by the paper. This section is persuasive as these examples are utilised to further prove the argument that the paper supports.
- **Results:** This section determines the outcome after data has been collected, analysed and measured, where the outcome can be utilised to confirm a theory or determine the effectiveness of the paper's solution. This section is written in an objective manner as it handles the outcome of objective, measured data.
- **Discussion:** This section analyses all the information that was collected throughout the paper and determines how it impacts what the paper is attempting to prove. It can consist of quantitative data that can be utilised as evidence to prove the paper's argument or qualitative data that suggests a shift towards a trend that the paper has identified. Therefore, this section is persuasive as the paper's researchers utilise the information that they identified to answer their research objectives and questions.

- **Future directions:** This section outlines events that will happen after the paper has been published, which can include likely trends identified from the data collected and other sources as well as additional contributions that the research team of the paper will make with their developed system or framework. Therefore, this section can be objective if it's based on the system that the paper's team will expand on later or persuasive if it's a solution that they the paper suggests.
- **Conclusions:** This section completes the paper by summarising the results and outcome of the contributions made by the paper. Therefore, this section is persuasive, because it's utilised to confirm the paper's findings.
- **References:** This section indicates all the sources utilised by the paper and where it can be located. This section is objective as it showcases all the sources that have impacted and were used in the paper.

3.2. Research paper overview and section structure

Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) AI-augmented CI/CD pipeline with anomaly detection and self-healing mechanisms. This framework utilises unsupervised learning techniques to detect anomalies by combining autoencoders, a deep neural network that reconstructs normal pipeline telemetry where anomalies are detected when reconstruction errors exceed a threshold, and isolation forests, a tree-based ensemble method that isolates anomalies by partitioning feature space at random. When an anomaly is detected a root cause analysis is performed where an LLM interprets the context of the failures and the issues that are associated with it. Moreover, these failures can either be fixed by the framework itself or require human intervention depending on the confidence scored assigned to the detected error.

Pareek (Pareek, 2026) argues that Explainable Artificial Intelligence (XAI) should be a requirement when adopting AI-driven software testing where the AI system can explain why certain decisions should be made, which can enhance trust, accountability and decision-making.

Widyastomo et al (Widyastomo and Lefaan, 2025) explores the challenges and solutions when implementing DevOps with CI/CD in complex, safety-critical environments. Additionally, a DevOps maturity assessment framework will be utilised to identify advantages and areas of improvement when implementing DevOps in embedded safety-critical environments.

Jain (Jain, 2023) discusses the implementation of continuous testing where automatic tests are performed throughout the software development lifecycle to prevent defects from reaching deployment.

Katapara et al (Katapara and Sharma, 2025) analyse when DevOps principles, like CI/CD, automated testing and deployment strategies, should be altered based on the constraints of embedded systems.

Sundara Raju et al (Sundara Raju, Yie Leong and Kumar Subramaniam, 2026) discuss about DevSecSafOps for the purpose of addressing both security and safety in complex systems through an integrated manner.

Kakarla et al (Kakarla and Sannareddy, 2024) creates an AI-driven DevOps Control Plane that is logically centralized and operationally federated, which enables enterprises to maintain heterogeneous CI/CD implementations while intelligence and governance remain consistent. This framework values safety over automation.

There are similarities and differences between the structure and content of each section mentioned in 3.1, which include:

- The structure of the introduction section remains consistent between each research paper except for Sundara Raju's paper that doesn't have a specified introduction section. Moreover, Sundara Raju's paper primarily utilises unique sections that are less board and more specific to the content of the section in comparison to the other research paper sections, such as the sections called "Implementation techniques for software safety testing" and "Evolution of DevOps".
- Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) and Kakarla et al (Kakarla and Sannareddy, 2024) papers have a related work section that mentions similar technologies that they utilise in their frameworks, however Kakarla et al primarily focusses on limitations and gaps that they have identified as their section also acts like a background section to provide further context for the justification of their framework.
- Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) and Kakarla et al (Kakarla and Sannareddy, 2024) utilise a proposed architecture/pipeline section that defines how each framework functions.
- Widyastomo et al (Widyastomo and Lefaan, 2025) differs from other research papers by combining both results and the discussion section into a single section called "Results and Discussion".

3.3. Identification and evaluation of research methodologies

Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) utilises a mixed methods approach where quantitative data are utilised as metrics, like Self-healing success rate and mean time to detection, to determine the performance of their proposed framework and a qualitative analysis is performed to determine the framework's effectiveness at handling repeating issues over a long period of time. Additionally, quantitative data is collected from three datasets that holds six months of data, such as build logs, test results and deployment records, that are sampled from industry partners in finance, e-commerce and SaaS, which will be used to detect anomalies. Public CI/CD failure benchmark from LogSage, comprising of 367 GitHub CI/CD failures, is also collected as it enables the Vasiliev et al to make comparisons with the root cause analysis accuracy with other root cause analysis approaches. In addition, known failure types, such as compilation errors, test failures and infrastructure timeouts can be utilised to measure remediation success rates. Moreover, the entire framework is deployed on key Mora metrics over 6 months, and the improvement is compared relative to the established baseline.

Pareek (Pareek, 2026) performs an interpretivist approach, because the paper is primarily conceptual by utilising publicly available sources and practical outcomes like system performance and developer feedback are unexplored in the paper. Therefore, no data collection or sampling methods were applied, and only other academic sources are utilised to explain and promote the concept of XAI.

Widyastomo et al (Widyastomo and Lefaan, 2025) utilises a case study methodology where qualitative data will be collected from interviews that are sampled from key stakeholders, including software developers, hardware engineers and operations team, and quantitative data, like performance metrics and the adoption of DevOps, is collected from surveys. In addition, simulation-based tests and real-world deployments are used to benchmark system performance, mostly focussing on resource usage and response time.

Jain (Jain, 2023) utilises an interpretivist approach through the use of case studies as qualitative data to prove the effectiveness of continuous testing in real-world scenarios. Moreover, the paper primarily focuses on defining the concept of continuous testing and the theoretical benefits, challenges and solutions associated with it, so no data collection or sampling methods are used.

Katapara et al (Katapara and Sharma, 2025) utilises an interpretivist approach where qualitative empirical and technical evidence was collected from other research papers and industry whitepapers. These documents consist of various types of academic resources related to embedded DevOps such as academic investigations, industry case studies, architectural patterns and toolchain evaluations. This paper and its chosen sources were utilised to create a grounded synthesis that provides practitioners with verified insights and other researchers with an agenda that can be used for future investigations.

Sundara Raju et al (Sundara Raju, Yie Leong and Kumar Subramaniam, 2026) utilises an interpretivist approach as no experiments or metrics were collected and only conceptual explanations of DevSecSafOps and its associated tools are used. Additionally, case studies were utilised to indicate the impact of safety incidents that could have been prevented with in-depth testing.

Kakarla et al (Kakarla and Sannareddy, 2024) utilises a positivist approach where its framework is evaluated using operational and governance metrics, that reflects real-world DevOps and SRE priorities, in a before-and-after comparative analysis across multiple, enterprise scale CI/CD environments. Additional quantitative data that will be used to determine the impact of the framework consists of data like mean time to detection, mean time to recovery and policy deviation rate.

3.4. Analysis of research findings and how it meets research objectives

Findings from Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) that contribute to each research objective and question include:

- **RO1:** In order for the framework to remain compliant to regulatory requirements, remediation actions are constrained to machine-readable policies to prevent automation from violating safety regulations. Additionally, all remediation actions, anomaly detections and root cause hypothesis have to be logged for post-incident analysis and high-risk actions must be approved by humans before it is performed. Moreover, this framework should be trained in a sandboxed pipeline environment before its utilised to make real decisions to reduce the likelihood of policy constraint violations indicated in the reward function.
- **RO2:** This framework is able to detect failures associated with tests and can attempt to resolve it by adjusting test configurations.
- **RO3:** This framework can perform remediation actions after analysing the root cause analysis, which includes retrying failed jobs with new parameters, patching dependency versions, rolling back artifact versions and creating pull requests that have code fixes.
- **RQ:** This framework can identify risks and anomalies and depending on the level of risk that a remediation action is categorised as result in either these remediation actions being performed or requiring human intervention.

Findings from Pareek (Pareek, 2026) that contribute to each research objective and question include:

- **RO1:** This paper explains that most AI models act as “black boxes” where it doesn’t provide sufficient reasoning for its actions, which could lead to incorrect actions being made without the developers realising it. Moreover, it states that there aren’t many approaches that utilise frameworks that combine compliance, auditability, and real-time mitigation.
- **RO2:** XAI can assist in automated testing by enabling developers to make decisions based on risk assessments with historical defect trends provided by the XAI, where these decisions can optimise test selections, reducing the amount of test cycles.
- **RO3:** XAI contributes to fail-safe mechanisms through automated testing that strengthens its compliance to regulations, which can reduce the likelihood of issues occurring.
- **RQ:** XAI can be utilised to improve trust in AI systems by enabling the system to meet regulatory requirements, allowing developers to understand the logic behind its decisions and suggestions in order to solve test issues in shorter test cycles than without using AI-driven technology.

Findings from Widyastomo et al (Widyastomo and Lefaan, 2025) that contribute to each research objective and question include:

- **RO1:** The main challenge when implementing CI/CD form embedded systems is the system’s limited computational resources and the real-time performance requirements requiring the system to respond to environmental changes where CI/CD could introduce delays, impacting the system’s performance. In addition, the complexity of embedded systems due to the various hardware components that require specialised configurations make it difficult to implement automated testing with conventional DevOps tools. If failures occur, it can result in severe consequences for safety critical industries like aerospace, healthcare and automotive.
- **RO2:** The paper suggests that customised frameworks should be developed to ensure that safety assurance processes occur in the CI/CD and tests should be performed in simulations to mimic the embedded system’s environment before changes are deployed to the real systems.
- **RO3:** The paper suggests that deep learning techniques can be valuable to these embedded systems in order to detect network attacks and respond to these attacks by reducing or preventing the negative impact that these attacks can have on the system.
- **RQ:** Traditional DevOps aren’t effective for embedded systems due to its unique characteristics; therefore, a customised framework of DevOps should be utilised to identify issues before deployment through CI/CD.

Findings from Jain (Jain, 2023) that contribute to each research objective and question include:

- **RO1:** Discrepancies that occur between test and production environments and are not discovered can create issues to the system when these errors are deployed, therefore, Infrastructure as Code (IaC) and containerisation should be utilised to find these discrepancies.
- **RO2:** A shift-left testing approach enables testing to occur in the beginning of the development cycle where issues can be caught before deployment. A shift-right approach focuses on real-world monitoring and A/B testing to ensure that system performance is stable and efficient.

- **RO3:** Real-time data can be collected post deployment in order to validate system's performance and rollback mechanisms if applicable.
- **RQ:** Automated testing throughout before deployment by using continuous testing can reduce the likelihood of issues being pushed in deployment by having more opportunities where issues can be detected than performing one test before deployment.

Findings from Katapara et al (Katapara and Sharma, 2025) that contribute to each research objective and question include:

- **RO1:** The are barriers to adopting DevOps for embedded systems such as CI failures when there were delays in the integration of software or hardware as well as flakiness when utilising Hardware-in-the-Loop (HIL) setups. Simulators are recommended to act as a bridging mechanism between DevOps and the embedded systems.
- **RO2:** Simulation-based testing enables tests to be performed separately from the physical system, allowing for improved reproducibility.
- **RO3:** The paper identified the difficulty of continuous development in an embedded system due to complexity of Over-The-Air (OTA), compliance requirements and rollback limitations.
- **RQ:** Conducting tests through simulations before deployment enables tests to be performed more efficiently than utilising the actual system.

Findings from Sundara Raju et al (Sundara Raju, Yie Leong and Kumar Subramaniam, 2026) that contribute to each research objective and question include:

- **RO1:** Safety-critical software like embedded systems have strict latency and reaction time requirements that can be difficult for developers to perform, because it's challenging to replicate every real-world scenario when performing real-time testing and performance analysis. Additionally, the integration of complex technology like AI, cloud computing and networked devices makes it difficult to determine every potential failure, which requires extensive testing that may not always be possible to achieve.
- **RO2:** Tools like Jenkins, GitLab CI/CD and Azure DevOps enables automatic test execution and scanning throughout software development. Moreover, tools like Coverity, SonarQube and PVS-Studio can be utilised to detect critical defects and security vulnerabilities through static code analysis.
- **RO3:** Additional techniques can be implemented to assist with automated regression testing to verify safety-critical components regularly, machine learning models that can predict potential failures before deployment and an AI-based anomaly detection system to detect run-time safety issues.
- **RQ:** This paper assists with confirming the effectiveness of testing and automating techniques displayed in the previous research papers like the machine learning elements in from Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) and testing tools from Katapara et al (Katapara and Sharma, 2025).

Findings from Kakarla et al (Kakarla and Sannareddy, 2024) that contribute to each research objective and question include:

- **RO1:** Failures that are deployed could only be discovered after impact.
- **RO2:** The framework has the ability to build test outcomes as well as adjust the test's scope or deployment. Moreover, learning models that are used to

detect issues can be updated to adopt to new issues through outcome feedback and human decisions.

- **RO3:** The framework contains fail-safe mechanisms such as using learning models to detect anomalies and failure trends.
- **RQ:** This framework is similar to the framework proposed by Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) that can identify anomalies and perform remediation actions with an adopting learning model.

3.5. Relationships and gaps between each research paper

Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025), Pareek (Pareek, 2026) and Kakarla et al (Kakarla and Sannareddy, 2024) agree on the importance of the explainability in an automated, AI-driven system with regards to regulatory compliance and auditability. Moreover, the frameworks from Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) and Kakarla et al (Kakarla and Sannareddy, 2024) extend the XAI techniques discussed by Pareek (Pareek, 2026) by implementing it in an actual system where Pareek (Pareek, 2026) only used conceptual information and did not utilise real-world examples. Moreover, Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025), Pareek (Pareek, 2026) and Kakarla et al (Kakarla and Sannareddy, 2024) also agree on the effectiveness of anomaly detection, rollback mechanisms and reinforcement learning techniques where systems are able to detect issues and respond accordingly, which can improve testing effectiveness. Additionally, Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) extends the self-healing tests that Jain (Jain, 2023) will perform in the future as the framework already utilises self-healing mechanisms.

However, there are gaps resented by Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) such as their framework not being tested in safety-critical sectors like healthcare, where the paper clarifies that additional verification mechanisms are most likely necessary, the paper not formally verifying that the AI policies satisfy safety properties before deployment in regulated industries and will be done in future research as well as the need for historical data to train the framework before it can be deployed. Additionally, both Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) and Kakarla et al (Kakarla and Sannareddy, 2024) don't mention the possibility that developers might misunderstand the root cause analysis when the framework requires human intervention as well as the potential for distrust from the developers on the reliability of AI systems that can complicate its deployment, which is discussed by Pareek (Pareek, 2026) Widyastomo et al (Widyastomo and Lefaan, 2025) mentions that CI/CD must also take into consideration the hardware constraints of embedded systems, which is not indicated from the frameworks from Vasiliev et al (Vasiliev, James Morrison and Zhang, 2025) and Kakarla et al (Kakarla and Sannareddy, 2024).

In addition, Pareek (Pareek, 2026) Jain (Jain, 2023) and Sundara Raju et al (Sundara Raju, Yie Leong and Kumar Subramaniam, 2026) don't perform any experimentations and Pareek (Pareek, 2026) doesn't showcase any case studies where XAI techniques are implemented in real-world scenarios as the paper is only conceptual. Pareek (Pareek, 2026) also mentions the possibility that unique solutions could have been overlooked as they only utilised recent or publicly available sources. Katapara et al (Katapara and Sharma, 2025) agrees with Widyastomo et al (Widyastomo and Lefaan, 2025) of the CI/CD challenges when implementing it for safety-critical embedded systems such as the integration of various unique characteristics from the hardware and software that makes it difficult for developers to prepare for every potential failure. Moreover, the

tools suggested by Katapara et al (Katapara and Sharma, 2025) were utilised by the sources indicated by Widyastomo et al (Widyastomo and Lefaan, 2025) such as Azure DevOps for automated testing and test execution.

4. Discussion

There are various issues and concerns when implementing traditional CI/CD principles in safety-critical embedded systems such as the overall complexity of the system due to the characteristics of the hardware and software utilised and how it interacts with each other, safety and regulatory compliances that must be followed and ensuring that the system meet strict performance criteria to prevent safety incidents. The information that has been collected throughout the literature review indicates challenges that can be solved by implementing specialised CI/CD pipelines that reduce the issues associated with embedded systems while providing the benefits of traditional CI/CD pipelines.

Challenges with automated deployment can include issues that aren't detected until after the negative impact has occurred, implementing XAI where the developers don't fully understand the explanation provided and allow the automated system to deploy issues without realising, the inability to fully test all possible errors in an embedded system without certain techniques and the possibility that deployments can decrease the performance of the system that can create dangerous situations where strict reaction time requirements must be met. There are various automated testing techniques that can be utilised for embedded systems such as simulation-based tests that mimic the embedded system and its environment that enables testing to be performed before deployment, tools like Azure DevOps enables automatic test execution and scanning throughout software development, customised frameworks that optimises testing for specifically embedded systems and XAI principles that enables developers to shorten test cycles. Moreover, automated fail-safe mechanisms and rollback features can be provided through AI-driven frameworks that are trained on historic failures and retrained to handle new failures that can occur while also providing remediation actions or require human intervention based on how much autonomy that developers give to the AI-driven system.

Therefore, CI/CD principles can successfully be implemented in safety-critical system and prevent deployment risks with rigorous testing throughout the CI/CD processes and simulations as well as machine learning techniques that can detect and remove anomalies before issues have an impact on the system.

5. Conclusions

Traditional CI/CD pipelines are not suitable for safety-critical systems due to the complexity of integrating various hardware and software together while having remain compliant with safety regulations. Therefore, customised CI/CD pipelines should be adopted for safety-critical embedded systems to handle the challenges associated with these systems through automated testing through simulations and automated anomaly detection and remediation with an AI-driven learning model that can adapt to new issues that can be present.

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

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